CSC 100 — Introduction to Programming in Python (Undergraduate)

Course Details

- Start & end: May 22 to August 28, 2026
- Instructor: Marcus Birkenkrahe, PhD
- Format: 2 x 75-minute sessions per week
- Date & time: Friday, 6 pm – 7:15 pm PDT
- Asynchronous: 75 min (optional DataCamp platform)

Course Description

Introduction to programming using Python. Emphasis on problem-solving, program design, and readable code. Students learn to construct programs step-by-step and develop computational thinking through guided practice and short projects.

Learning Objectives

- Write, run, and debug Python programs
- Use variables, conditionals, loops, and functions
- Work with lists and dictionaries
- Decompose problems into smaller steps
- Read and explain code clearly
- Develop small, complete programs independently

Course Materials

Required

- *Think Python (2nd ed.)* — Allen B. Downey

Provided

- Colab notebooks (in-class)
- Practice files
- DataCamp (practice only)
- Gemini (AI tutor, see policy)

Weekly Schedule

Week	Topic	In-Class Focus	Reading	Project
1	Introduction	Python, expressions	Ch 1	
2	Variables & Types	Assignment, types	Ch 2	

Week	Topic	In-Class Focus	Reading	Project
3	Conditionals	if/else	Ch 5	P1 start
4	Functions I	Defining functions	Ch 3	
5	Functions II	Parameters, return	Ch 6	P1 due
6	Loops	while, for	Ch 7	P2 start
7	Loop Patterns	Accumulation	Ch 8	
8	Integration	Combined problems	Review	P2 due
9	Strings	Indexing, slicing	Ch 9	P3 start
10	Lists	Traversal	Ch 10	
11	Dictionaries	Key-value	Ch 11	P3 due
12	Files	Read/write	Ch 14	P4 start
13	Problem Solving	Multi-step	Select	
14	Project Work	Debugging	Review	
15	Final Review	Synthesis	Review	P4 due

DataCamp (asynchronous)

Week	Topic	Module
1–2	Basics	Introduction to Python
3–5	Control flow	Intermediate Python (core sections)
6–7	Loops	Intermediate Python
8–11	Structures	Lists, dictionaries modules
12+	Files/data	Importing Data (optional)

Grading

Component	Weight
DataCamp lessons	20%
Weekly exercises	40%
Projects (4)	20%
Weekly quizzes	20%

AI Usage Policy

Students may use AI tools to:

- Ask for explanations
- Debug code

Students must:

- Explain their code
- Modify their solutions when asked

Instructional Approach

- Code-first learning
- Small, frequent exercises
- Incremental projects

Verification

- In-class coding
- Short quizzes
- Project variation and explanation

CPU policies

For policy matters and procedures, especially concerning attendance, grading tables, withdrawals etc. please consult the Catholic Polytechnic University's CATALOG OF COURSES (2025-2026).

Author: Catholic Polytechnic University
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